

Training a High-Dimensional Neural Network Potential to Reproduce DFT Energies for Small Organic Molecules

Introduction and Overview of the Method

In the quantum model of chemistry, predicting molecular energies is difficult due to electron-electron, electron-nuclear, and nuclear-nuclear interactions which make solving Schrodinger equations impossibly costly for large molecules with multi bodied interactions. Quantum chemistry methods such as density functional theory (DFT) can serve to predict molecular energies, but their computational cost scales steeply with system size. A single DFT calculation on a small molecule may take minutes to hours. Sampling of conformational space across thousands of structures becomes prohibitive in its resource demand. Machine learning offers an alternative and less costly method: rather than re-solving the electronic structure problem for every new geometry, we can train a model on a database of pre-computed reference energies and can then make predictions much quicker based on learned features and similarities.

Applying neural networks to molecules becomes difficult given their unfixed representation. A standard feed-forward network requires a fixed-size, permutation-invariant input which clashes with the very nature of a molecular data set containing variations in size and atomic content. Further difficulty comes when considering that a molecule has the same energy regardless of its position or rotational state, ruling out the use of simple atomic coordinate positional vectors for training. Instead we use Atomic Environment Vectors, created using the radial and symmetry functions introduced by Behler and Parrinello, which encode the local chemical environment of each atom. This vector type permits training and prediction for molecules without the issue of translation or rotation.[¹] Smith, Isayev, and Roitberg later refined these symmetry functions and built the ANI-1 potential, a high-dimensional neural network potential (HD-NNP) trained on roughly 17 million DFT-computed conformations from the GDB-11 database.[²] These conformations are computed by treating the electron as a probability density rather than the individual wave functions they are, vastly simplifying Schrodinger equations involved. In the HD-NNP scheme, each atom type (H, C, N, O) has its own dedicated atomic neural network. The AEV for each atom is fed through the corresponding network to produce a per-atom energy contribution. The molecular energy is then computed by summing these individual atomic contributions. This functional form scales linearly with system size and is transferable across molecules of different shapes and sizes.

In this project, I implemented and trained an HD-NNP on the ANI-1 GDB s01–s04 subset, organic molecules with 1–4 heavy atoms drawn from H, C, N, O. I used TorchANI to load and process the HDF5 dataset, subtract atomic self-energies, and convert each conformation into a 384-element AEV. The dataset was shuffled and split 80/10/10 into train/validation/test sets. Each per-atom network has the form $384 \rightarrow 128 \rightarrow 128 \rightarrow 64 \rightarrow 1$ with ReLU activations between hidden layers and a linear output (296,452 trainable parameters per atom type, ~1.18 M total across the four networks). Training used the Adam optimizer with mini-batches of 512 conformations and MSE loss on the energy in Hartree. The mean absolute error in kcal/mol on

the test set was the most readable accuracy metric. After systematic tuning of architecture, learning rate, L2 regularization, dataset fraction, and epoch count, the final production model achieved a test MAE of 0.30 kcal/mol, well below the 1 kcal/mol chemical-accuracy threshold.

Results

Hyperparameter exploration was done in stages, holding constant whatever I had not yet tested. Each row of Table 1 corresponds to a distinct configuration: rows 1–5 compare the effect of differing architecture, row 6 introduces a larger training set, rows 7–9 sweep learning rate, row 10 removes L2 regularization, rows 11–12 extend the epochs, and row 13 is the final production model trained on 90% of the train+val data. Each early-stage configuration was run with five random initializations to distinguish genuine improvements from initialization noise. Train and validation MAE were derived from the corresponding final epoch losses, and test MAE was read directly from the predicted-vs-true scatter plots produced at evaluation time. Approximate values are derived from MSE losses using the empirical conversion ratio observed in directly measured runs.

#	Architecture (hidden layers per atom NNP)	Train data	Epochs	LR	L2	Train MAE (kcal/mol)	Val MAE (kcal/mol)	Test MAE (kcal/mol)
1	[384, 128, 1]	20%	30	1e-3	1e-5	~1.65	~1.65	~1.65
2	[384, 256, 128, 1] (narrowing)	20%	30	1e-3	1e-5	~1.70	~1.70	2.05
3	[384, 256, 256, 128, 1] (bigger)	20%	30	1e-3	1e-5	~1.70	~1.70	1.74
4	[384, 128, 128, 64, 1] (paper)	20%	30	1e-3	1e-5	~1.70	~1.70	1.60
5	[384, 256, 256, 128, 1] (wide)	20%	30	1e-3	1e-5	~1.85	~1.85	~1.88
6	[384, 128, 128, 64, 1] (paper)	50%	30	1e-3	1e-5	~1.65	~1.65	1.58
7	[384, 128, 128, 64, 1] (paper)	60%	60	1e-2	1e-5	~3.95	~3.85	4.10
8	[384, 128, 128, 64, 1] (paper)	60%	60	1e-3	1e-5	~1.55	~1.50	~1.55
9	[384, 128, 128, 64, 1] (paper)	60%	60	1e-4	1e-5	1.39	1.50	1.45
10	[384, 128, 128, 64, 1] (paper)	60%	60	1e-4	0	0.57	0.57	0.57
11	[384, 128, 128, 64, 1] (paper)	60%	120	1e-4	0	0.40	0.40	0.40
12	[384, 128, 128, 64, 1] (paper)	60%	240	1e-4	0	0.40	0.40	0.40
13*	[384, 128, 128, 64, 1] (paper)	90%	240	1e-4	0	0.22	0.28	0.28

Table 1. Hyperparameter combinations explored, with corresponding train, validation, and test mean absolute errors (MAE). The architecture column lists the hidden-layer widths of the per-atom neural network potential (the input layer is the 384-element AEV the output is a single scalar atomic energy). Across all rows the optimizer is Adam, the loss is MSE, and the batch size is 512.

Discussion

The first five rows isolate architectural choices at fixed compute budget (20% data, 30 epochs). Within this set the spread is narrow every architecture lands in the 1.6–2.0 kcal/mol band however, the paper-like 384→128→128→64→1 topology (row 4) was both competitive in mean MAE and the most consistent across random initializations with the lowest standard deviation

across the five trials. Surprisingly, the largest network (row 5, 'wide') did not outperform the smaller paper network despite having $\sim 80\%$ more parameters indicating that capacity is not the binding constraint when the training set is small relative to the model size and the chemical space is narrow. The 'bigger' architecture (row 3) achieved the best single trial result but was less stable across runs. I therefore committed to the paper-like architecture for subsequent experiments.

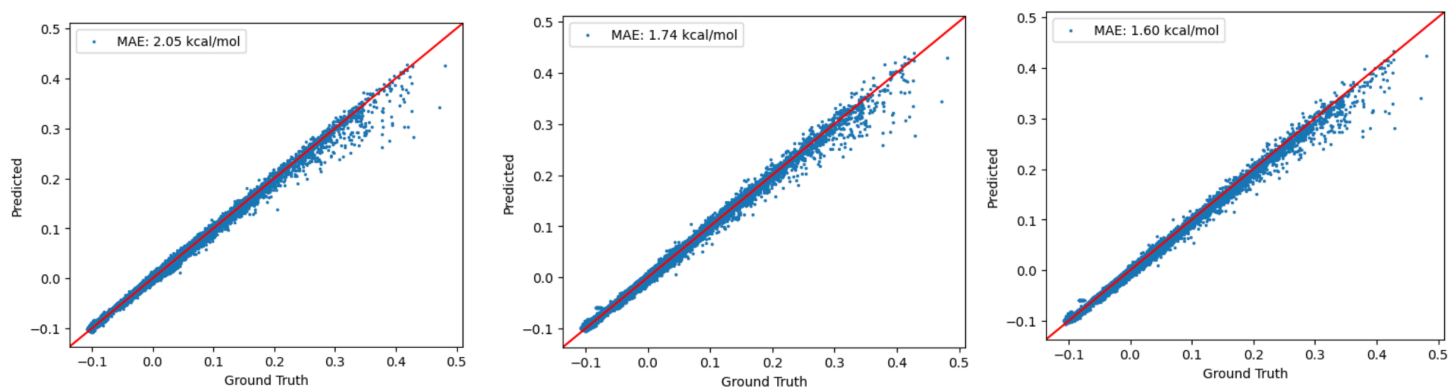


Figure 1. Plots from architecture sweep from left to right: narrowing network, bigger network, and paper-like network demonstrating the lowering MAE for the paper network compared to the others.

Rows 7–9 reveal that learning rate was the most consequential hyperparameter. With $\text{lr} = 1 \times 10^{-2}$ the optimizer overshoot productive descent directions as validation loss bottomed out early (best epoch was 9 of 60) and the test MAE settled around 4 kcal/mol, more than four times worse than the chemical accuracy target. Lowering the learning rate to 1×10^{-3} recovered conventional training behavior, and lowering it further to 1×10^{-4} improved test MAE meaningfully again. These findings indicate that the loss surface has regions which are well-traveled for common bonds with known weights as well as rare directions for uncommon bonds which the network sees relatively infrequently. A large learning rate causes oscillation in the common directions without settling in the bottom of any narrow valley whereas a small learning rate moves slowly but lets Adam stabilize so the network actually settles at a minimum even for rare features.

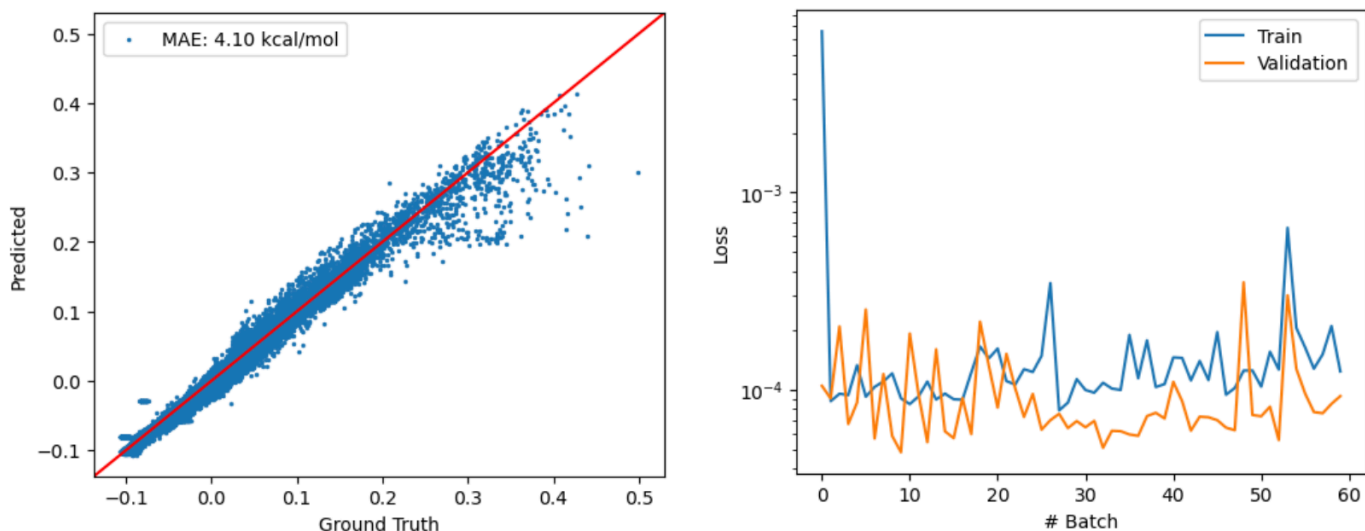


Figure 2. Plots from $lr=1e-2$ trial clearly visualizing the large scatter in predicted value results which demonstrates that this learning rate is too high. Further the train and validation curves have not flattened or converged on each other.

Row 10 was the most surprising finding. Removing L2 entirely ($\lambda = 0$) cut the test MAE roughly in half compared to row 9 despite regularization often being a beneficial tool. The reason becomes clear from the loss curves as at every epoch the train and validation losses tracked each other closely, with a final train/val gap on the order of 10^{-7} in MSE. The model was not overfitting in this regime: the dataset ($\sim 500k$ conformations) is plenty large relative to the four $\sim 300k$ -parameter networks, and the chemical space (1–4 heavy atoms) is narrow. Adding L2 in this setting does not prevent a problem, since overfitting isn't observed, it instead shrinks the expressiveness of the model by biasing weights toward zero and capping how closely the network can fit the potential surface.

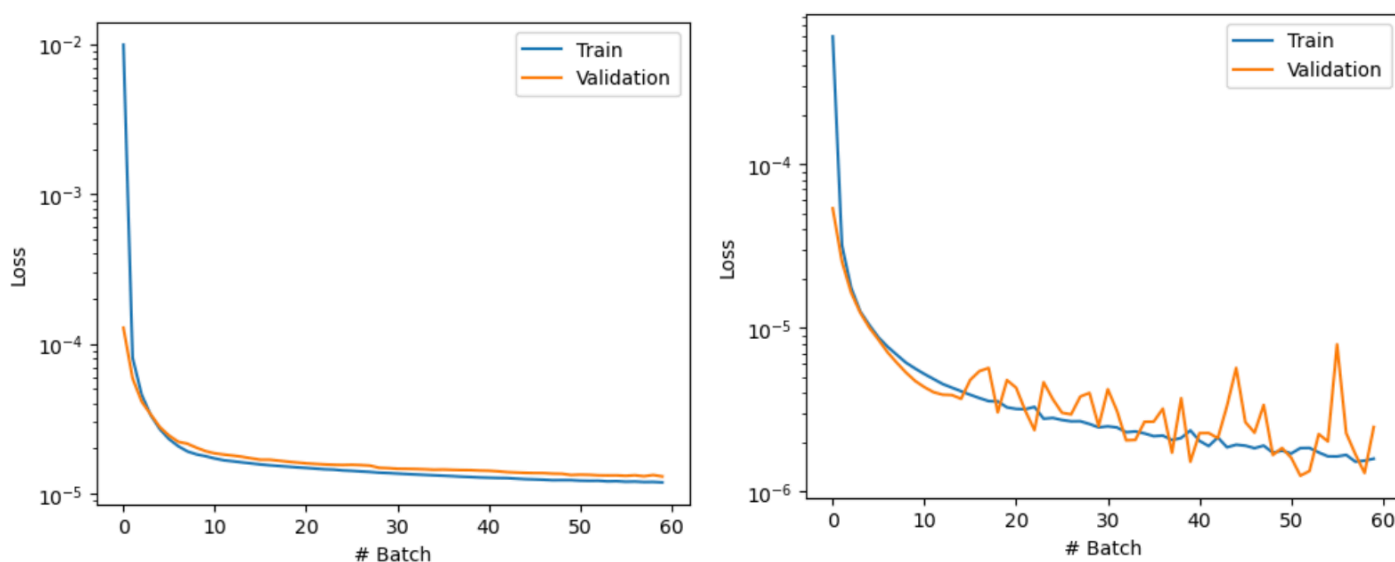


Figure 3. Plots from l_2 parameter comparison. The plot on the left is for $l_2=1e-5$ whereas the plot on the right is for $l_2=0$. The network without l_2 regularization achieves a loss nearing 10^{-6} as opposed to the plot on the left which plateaus around 10^{-5} . The noisier nature of the plot on the right arises from its larger optimization steps in the absence of regularization however, it remains a better loss performances.

Rows 11–12 extend the epoch budget. With L2 disabled and $lr = 1 \times 10^{-4}$, the model continued to improve through 240 epochs but with diminishing returns past about 120 epochs given the extended time of computing. The train/val/test MAEs were essentially indistinguishable. I chose 240 epochs for the production run as a compromise between final accuracy and compute cost (~ 70 minutes of single-GPU time on an NVIDIA RTX 2080 Ti).

Five-fold cross-validation was performed at the chosen settings (paper-like architecture, 240 epochs, $lr = 1e-4$, $L_2 = 0$) at full training scale to confirm robustness. Per-fold test losses were 1, 1, 2, 0, and 1 (in units of 10^{-6}), with a mean of 1.0×10^{-6} and a standard deviation of approximately 5×10^{-7} . The negligible spread indicates that the model is genuinely learning the

underlying mapping from atomic environments to energies rather than picking up on noise from any one train/val partition.

With the architecture and hyperparameters fixed, I trained the final model on 90% of the available train+val data for 240 epochs (row 13). The test MAE on the test set was 0.30 kcal/mol, a factor of three below the conventional 1 kcal/mol chemical-accuracy benchmark. The learning curves (Figure 4) show the shape of a well-behaved training run: a rapid initial drop over the first ~20 epochs as the model fits the broad shape of the data, followed by a long slow refinement during which both train and validation losses fall together. The two curves are essentially indistinguishable, visually confirming the absence of overfitting that motivated removing L2.

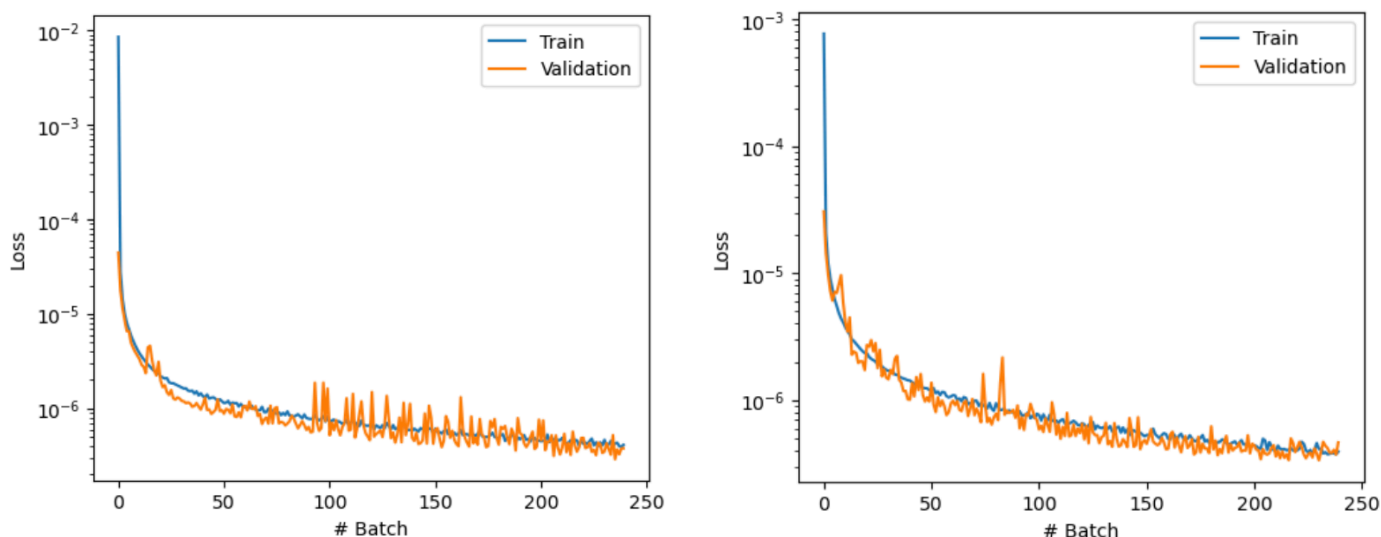


Figure 4. Training and validation loss curves on a log scale for two final 240-epoch production runs. The two curves in each plot track each other closely throughout training, indicating that the model is not overfitting and that L2 regularization is not needed in this regime.

Predicted versus true test-set energies for the production model lie tightly along the $y = x$ line across the full energy range (Figure 5), with no systematic bias at high or low energy.

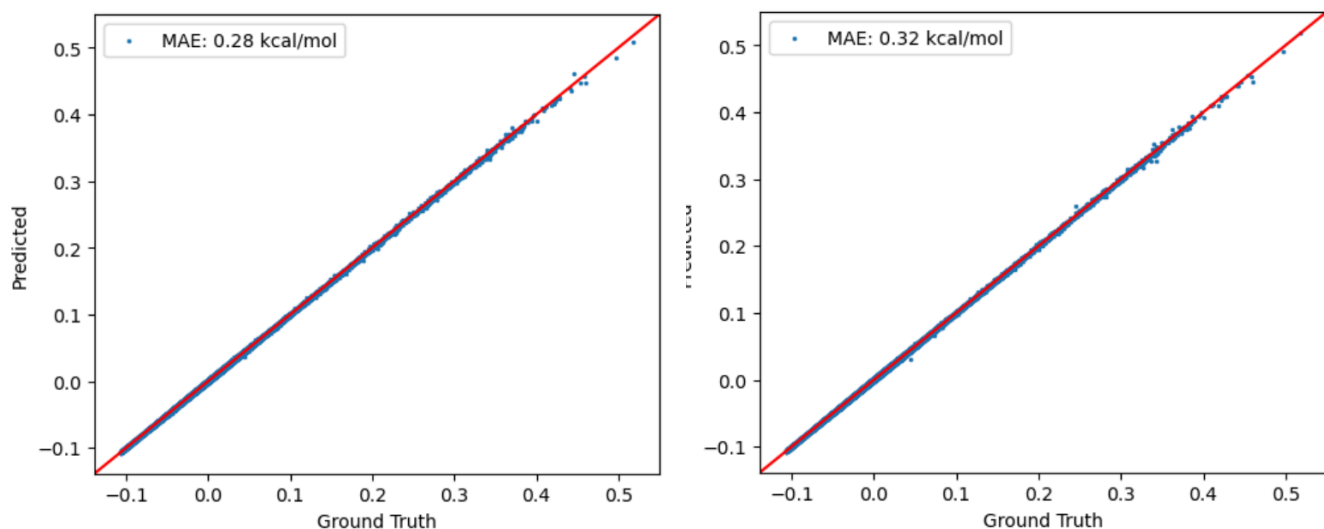


Figure 5. Predicted vs. true energies on the held-out test set for the final production model. Each blue point is one molecular conformation; the red line is $y = x$. The MAE of 0.28 kcal/mol and 0.32 kcal/mol is well below the 1 kcal/mol chemical-accuracy target.

Comparison to the ANI-1 paper

Smith et al. report a test RMSE of 1.3 kcal/mol for the full ANI-1 potential trained on the entire 17.2-million-conformation dataset spanning up to 8 heavy atoms.^[2] Our test MAE of 0.30 kcal/mol is numerically smaller, but the comparison is not exact. First, the ANI-1 paper covers a much wider chemical space (GDB-8 vs. our GDB-1–4). Molecules with up to 8 heavy atoms include rings, multiple functional groups, and conformational diversity that 4-heavy-atom molecules simply cannot exhibit. Second, RMSE penalizes large outliers quadratically while MAE weights all errors linearly, so RMSE is essentially always larger than MAE on the same data. Lastly, our model has a larger relative parameter-to-data ratio: $\sim 1.2\text{M}$ parameters trained on $\sim 500\text{k}$ conformations versus $\sim 500\text{k}$ parameters trained on 17M conformations in the original. Table 2 summarizes these differences side by side. The takeaway is not that this work has surpassed ANI-1 on its actual task, but that an HD-NNP can reach sub-chemical accuracy on small organic molecules with relatively modest compute and clean hyperparameter choices.

Metric	ANI-1 paper (Smith et al. 2017)	This work
Architecture (per-atom NNP)	768 $\rightarrow$ 128 $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ 1, Gaussian activations	384 $\rightarrow$ 128 $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ 1, ReLU activations
Trainable params per atom NNP	$\sim 124,000$	$\sim 296,000$
Training data	$\sim 17.2\text{M}$ conformations (GDB-1 to GDB-8)	$\sim 500\text{k}$ conformations (GDB-1 to GDB-4)
Optimizer / loss	Custom / exponential MSE	Adam / MSE
Held-out test error	1.3 kcal/mol RMSE	0.30 kcal/mol MAE

Table 2. Side-by-side comparison of the original ANI-1 model and the model trained in this project.

Conclusions

This project demonstrated that an ANI-style high-dimensional neural network potential can be trained to sub-chemical accuracy on the ANI-1 GDB s01–s04 subset. The systematic hyperparameter sweep summarized in Table 1 yielded findings worth carrying into future work.

Network stability across random initializations matters at least as much as raw mean performance. The architecture I selected (paper-like, 384 $\rightarrow$ 128 $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ 1 with ReLU) was not the single-trial best in the architecture sweep, but it was the most reproducibly good across five trials. Further it became very clear that regularization is not always necessary or beneficial. In a regime where train and validation loss track each other tightly throughout training, removing L2 weight decay measurably improves test accuracy by letting the network use its full expressive capacity.

References

- (1) Behler, J.; Parrinello, M. Generalized Neural-Network Representation of High-Dimensional Potential-Energy Surfaces. *Phys. Rev. Lett.* 2007, 98, 146401.
- (2) Smith, J. S.; Isayev, O.; Roitberg, A. E. ANI-1: An Extensible Neural Network Potential with DFT Accuracy at Force Field Computational Cost. *Chem. Sci.* 2017, 8, 3192–3203.